Book 3: Within-subject fine-grained analyses of the auditory superstitious-perception task

Auditory SP analyses

Contents

1 What this book is and how to read it

Books 1 and 2 worked at the group level. Book 1 asked how the experimenter's target / distractor labels relate to acoustic similarity between stimuli, and Book 2 asked how subjects' responses relate to those labels and to one another. The headline result of Book 2 was that group-level alignment is small but not zero ($d' \approx 0.09$ in the `full_sentence` block, $p \approx 0.006$), and that the cross-subject behavioural geometry has no clear cluster structure.

Book 3 zooms inside individual subjects. The questions are:

1. **Per-subject classification image (Ch. 22).** If we average the raw waveforms of every stimulus a subject called "target" and subtract the average waveform of those they called "distractor", what does the resulting per-subject *template* look like, and how aligned is it to the experimenter's true target-minus-distractor template (and to the original `wall.wav` sentence)?

2. **Within-subject internal consistency (Ch. 23).** Is each subject's behaviour reproducible across random halves of their trials, in either raw waveform or magnitude-spectrum space?

3. **Spectral logistic alignment (Ch. 24).** If we model each subject's responses as a linear logistic decision over 32 log-spaced magnitude-spectrum bands, how aligned is their weight vector to the analogous weight vector trained on the experimenter labels?

4. **Reminder-anchored and block-position effects (Ch. 25).** Every 15 trials, subjects heard a *reminder* replay of the `wall.wav` template. Do response, hit, and false-alarm rates depend on how many trials have elapsed since the last reminder, or on absolute position within the 150-trial block?

5. **Block-order and site (Ch. 26).** Does it matter whether a block was run first or second, and do Oberlin and UChicago participants differ?

6. **d' tertiles – who carries the signal (Ch. 27).** Split subjects within block by d' and by $|d'|$ and ask which sub-group the alignment effect lives in.

7. **Individually-significant subjects (Ch. 28).** A close look at the few subjects who survive a per-subject permutation test, including their questionnaire profile.

8. **Questionnaire correlates (Ch. 29).** Do BAIS_V (auditory imagery vividness), BAIS_C (control), TAS, LSHS, DES, FSS, VHQ, or Stanford sleepiness predict any per-subject signal metric?

Reading guide. Each chapter has four short sections: *plain-English intent*, what we do, the result you should look at, and how to interpret it. Every figure in this book is a per-subject summary; group-level numbers are reported in the text but the visual emphasis is always on within-subject distributions.

Reminder structure. Within each of the two 150-trial blocks the reminder screen fired before trials $15, 30, 45, \ldots, 135$. So for any trial in either block, two useful temporal covariates are `trials_since_reminder` $=$ `trial_index` mod $15 \in \{0, \ldots, 14\}$ and `pos_in_block` $=$ `trial_index` $\in \{0, \ldots, 149\}$.

A useful hierarchical framing. Throughout this book each per-subject statistic θ_s (e.g. a cosine alignment, a logistic slope) is implicitly viewed as $\theta_s \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ with μ_θ the population effect and σ_θ the between-subject variability. A non-zero μ_θ is what we mean by "the typical subject is doing the task"; a positive σ_θ even with $\mu_\theta = 0$ would mean "different subjects use systematically different but real strategies". We don't fit a full Bayesian hierarchical model here, but every group-level statistic in this book is the random-intercept version of the corresponding μ_θ.

2 Chapter 22 – Per-subject classification image

2.1 Plain-English intent

For each subject, build the raw audio template they implicitly endorsed by calling certain stimuli "target" and others "distractor". Compare it to (a) the experimenter's own target-minus-distractor template for the block, and (b) the original `wall.wav` clip the subject was told to listen for.

2.2 Method

For subject s in block b with response vector $y_s \in \{0, 1\}^{150}$ (1 = called target):

$$\begin{aligned} \mathrm{CI}_s &= \overline{w}_{y_{si}=1} - \overline{w}_{y_{si}=0}, \\ \mathrm{CI}_{\mathrm{true}} &= \overline{w}_{t_i=1} - \overline{w}_{t_i=0}, \end{aligned}$$

where w_i is the centred 2960-sample waveform of stimulus i and t_i is its experimenter label. We report two cosines per subject: $\cos(\mathrm{CI}_s, \mathrm{CI}_{\mathrm{true}})$ and $\cos(\mathrm{CI}_s, \texttt{wall.wav})$. The null is built per subject by permuting y_s across stimuli ($N = 1000$) and recomputing the cosine.

2.3 Result

Across both blocks the experimenter template $\mathrm{CI}_{\mathrm{true}}$ is itself only mildly aligned to `wall.wav` ($\cos \approx 0.42$): a target stimulus is acoustically a distorted, embedded copy of the sentence rather than a clean copy. At the per-subject level (Fig. **??**):

- `full_sentence`: mean $\cos(\mathrm{CI}_s, \mathrm{CI}_{\mathrm{true}}) = +0.034$, **77% of subjects above zero**, no individual subject permutation-significant at $p < 0.05$.
- `imagined_sentence`: mean $+0.014$, 58% above zero, one subject individually significant.

So the *group* signal is real (a binomial test on 20/26 positive gives $p < 0.01$ for `full_sentence`) but no individual subject's classification image is, on its own, reliably aligned. This is the characteristic "many small noisy effects, all in the same direction" pattern. Figure **??** shows the worst- and best-aligned subject in each block; even the best aligned CI_s does not look like a clean copy of $\mathrm{CI}_{\mathrm{true}}$.

2.4 How to interpret

Subjects appear to be *slightly* biased towards endorsing acoustically target-like stimuli, but the within-subject signal is roughly the same order of magnitude as the per-subject permutation null. What pushes the group result is the consistent *sign* of the cosine across subjects, not the magnitude in any one of them.

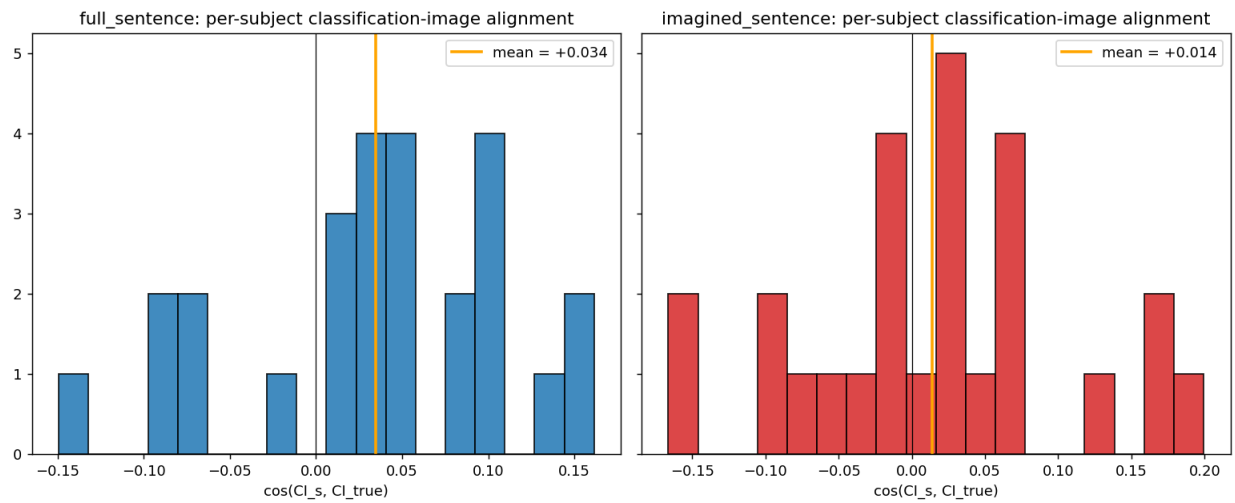

Figure 1: Per-subject cosine between the subject's classification image CI_s and the experimenter template $\mathrm{CI}_{\mathrm{true}}$. Each bin is one subject. Vertical orange line: group mean.

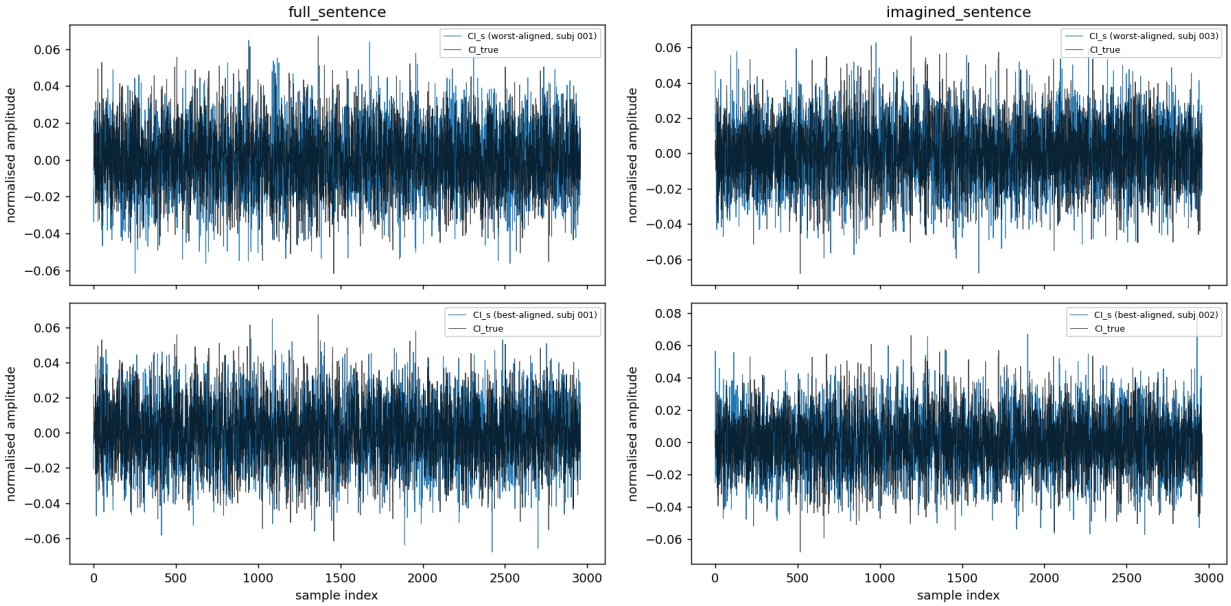

Figure 2: Worst-aligned (top row) and best-aligned (bottom row) subject's classification image (blue, normalised) overlaid on the experimenter template (black, normalised), per block.

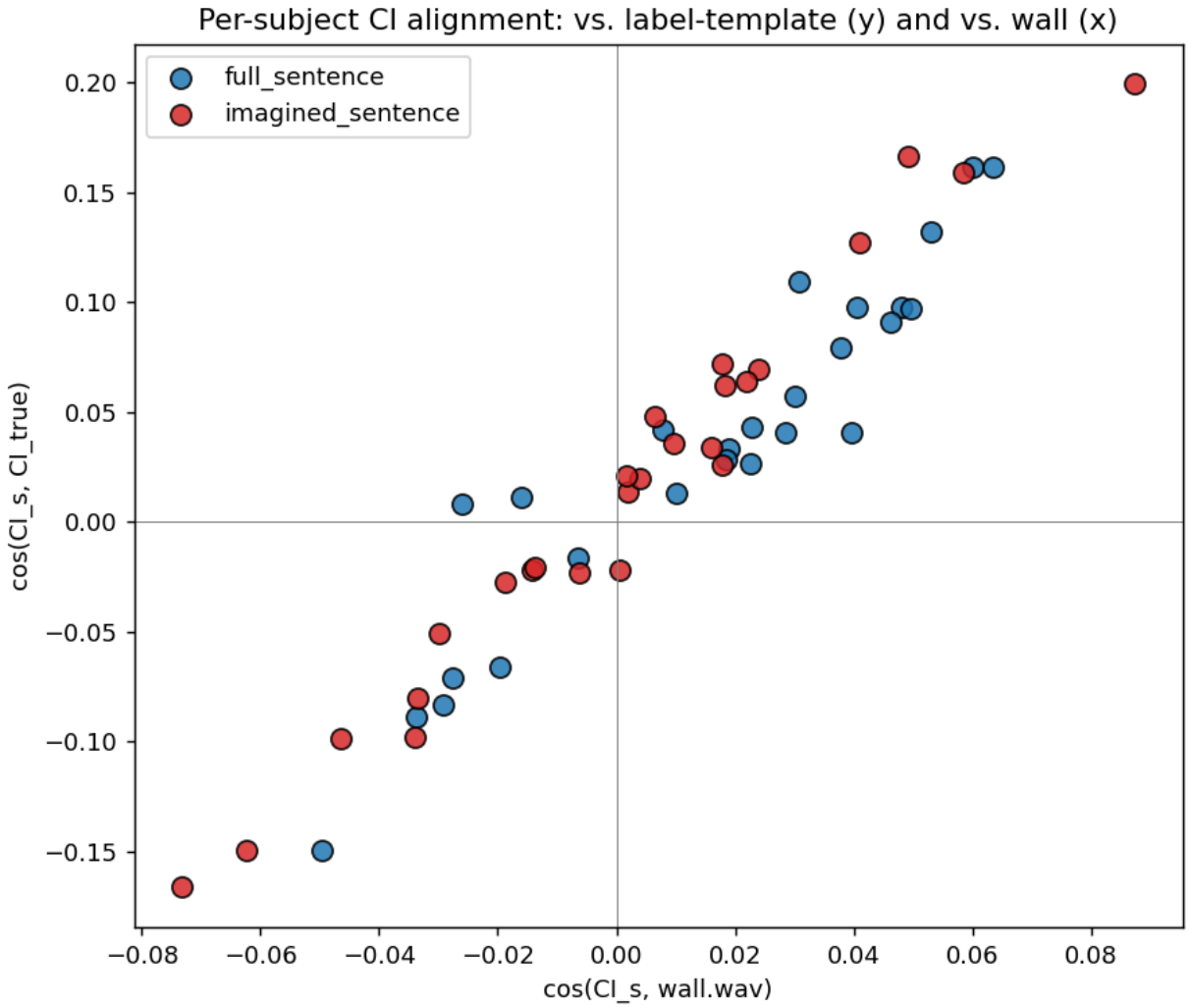

Figure 3: Per-subject classification-image cosines: y-axis vs $\mathrm{CI}_{\mathrm{true}}$, x-axis vs the original `wall.wav`. Subjects above the horizontal axis are biased towards the experimenter template; subjects to the right of the vertical axis are biased towards the raw sentence.

3 Chapter 23 – Within-subject split-half consistency

3.1 Plain-English intent

If a subject is using *any* consistent rule – experimenter-aligned or not – to decide which stimuli to call "target", then the classification image computed on a random half of their trials should look like the classification image computed on the other half. We test this in two feature spaces: raw centred waveform (2960-D) and the magnitude FFT (1481-D).

3.2 Method

For each subject, repeat 200 random splits of the 150 trials into halves A, B. Compute $\mathrm{CI}_A, \mathrm{CI}_B$ on each half and report the average Pearson correlation $\bar{r}_s = \mathbb{E}_{\text{splits}}\, \mathrm{corr}(\mathrm{CI}_A, \mathrm{CI}_B)$. A response-shuffled null gives the floor.

3.3 Result

Both feature spaces give $\bar{r}_s$ values indistinguishable from zero (group means in $[-0.007, +0.005]$) and indistinguishable from the shuffled null (Fig. **??**). Plotting $\bar{r}_s$ against the chapter-22 alignment cosine shows no relationship (Fig. **??**).

3.4 How to interpret

Two readings:

1. Subjects are *not* consistently using a rule that lives in raw waveform or magnitude-spectrum coordinates. Whatever consistency they have (chapter 22 says some sign-consistency exists) lives along the *single* direction defined by $\mathrm{CI}_{\text{true}}$, not along arbitrary directions in a $\sim$1500-D feature space. Estimating an arbitrary direction from 75 trials is statistically hopeless even for perfectly consistent rules; estimating the projection onto a known direction is fine.

2. This is therefore a *negative* result for "hidden idiosyncratic strategies in raw acoustics". If subjects each have their own private strategy in either feature space, the strategy is too weak to detect with 75 yes-trials and 75 no-trials per block.

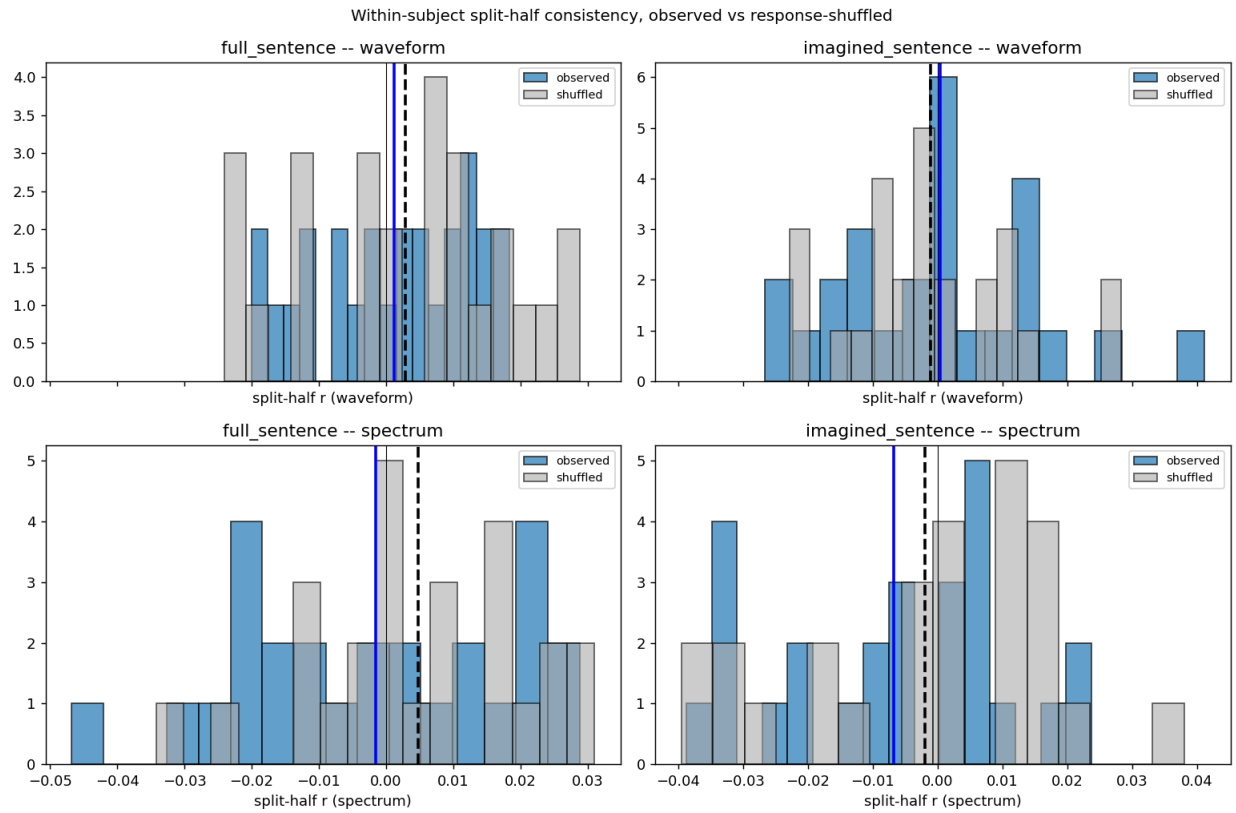

Figure 4: Per-subject split-half consistency $\bar{r}_s$ in waveform (top) and magnitude-spectrum (bottom) space, observed (blue) vs response-shuffled (gray), per block. Both observed distributions are indistinguishable from their shuffled controls.

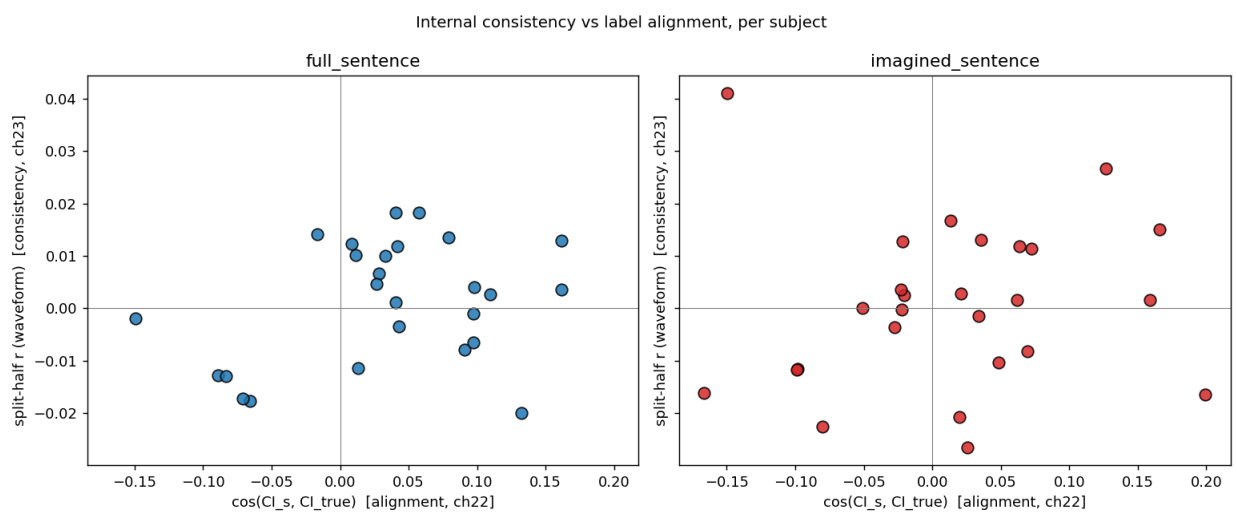

Figure 5: No relationship between within-subject consistency (y, ch. 23) and label-template alignment (x, ch. 22).

4 Chapter 24 – Per-subject spectral logistic regression

4.1 Plain-English intent

Reduce each stimulus to its magnitude-spectrum amplitude in 32 log-spaced frequency bands ($\sim$50 Hz to $\sim$3950 Hz), then fit a ridge logistic regression per subject: $P(\text{called_target}_i = 1) = \sigma(w_s^\top x_i + b_s)$. Compare the resulting weight vector w_s to the analogous weight vector w_{true} trained on the experimenter labels for the same block.

This is the same spirit as chapter 22, but in a much lower-dimensional basis where 75 yes-trials *is* enough to estimate a direction.

4.2 Method

Per stimulus, compute the magnitude FFT and average within each of 32 geometrically-spaced bands. Standardise across the 150 stimuli. Per-subject ridge logistic with $C = 0.5$ (sklearn, `liblinear`). w_{true} is the same model fit with y=experimenter label. Per-subject permutation null: shuffle y_s and refit ($N = 200$ shuffles).

4.3 Result

- `full_sentence`: mean $\cos(w_s, w_{\text{true}}) = +0.064$, 65% positive, **2/26 subjects individually** $p < 0.05$.
- `imagined_sentence`: mean +0.042, 54% positive, no subject individually significant.

Figures **??** and **??** show the distribution and the per-subject weight vectors sorted by alignment. In `full_sentence` the highest-aligned subjects' weight vectors visibly mirror the dominant low-frequency structure of w_{true}. Figure **??** shows that within `full_sentence` subjects with higher chapter-16 d' also tend to have larger $\cos(w_s, w_{\text{true}})$ – exactly what one would expect if the spectral-band logistic is capturing the same per-subject discrimination signal that SDT captures.

4.4 How to interpret

This is the strongest within-subject evidence in the book that individual participants are using the experimenter-defined acoustic information, not just at the group level. The effect is small in magnitude but consistent in direction, and the spectral basis is small enough that two individual subjects survive a per-subject permutation test.

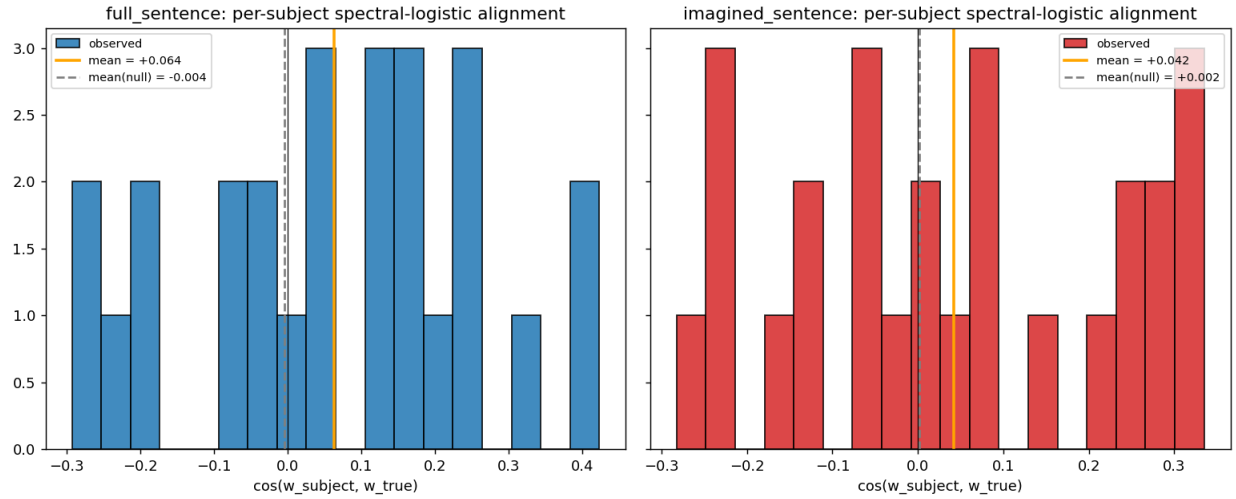

Figure 6: Per-subject cosine alignment between the subject's spectral logistic weight vector w_s and the experimenter weight vector w_{true}. Dashed gray line: subject-averaged null mean.

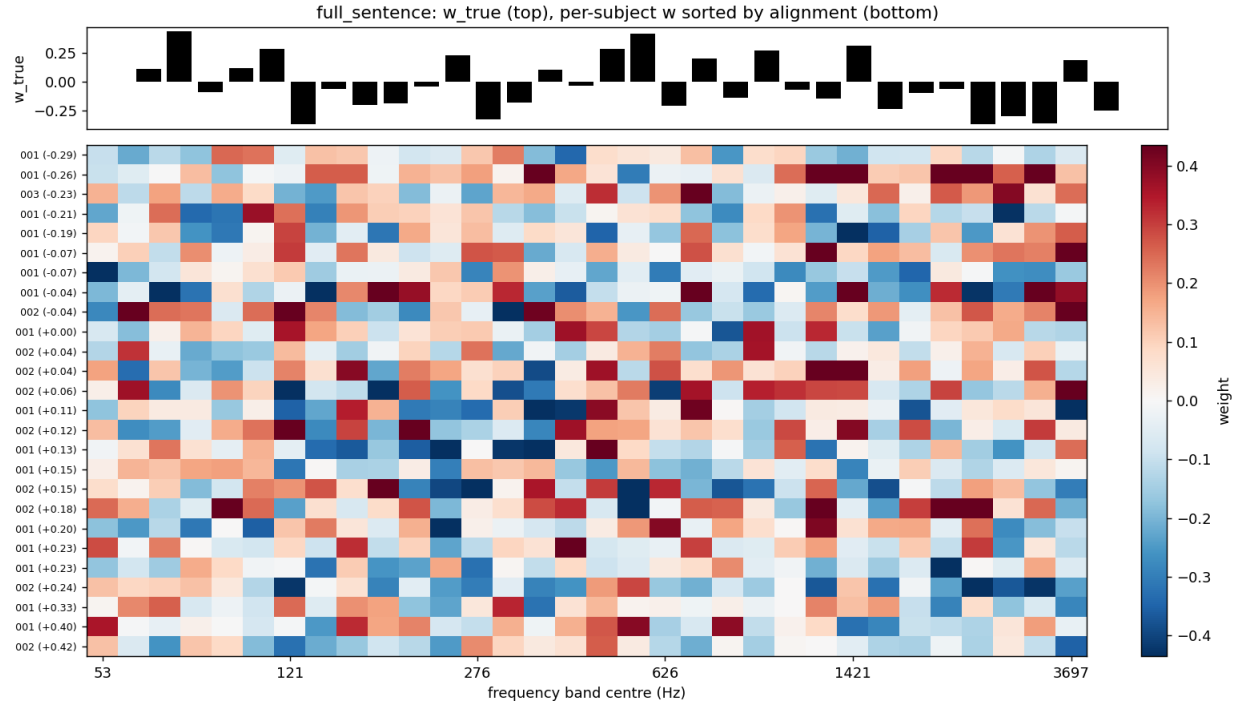

Figure 7: `full_sentence`: w_{true} across the 32 bands (top) and per-subject w_s rows (bottom), sorted by alignment cosine. x-axis tick labels: band centre frequency in Hz.

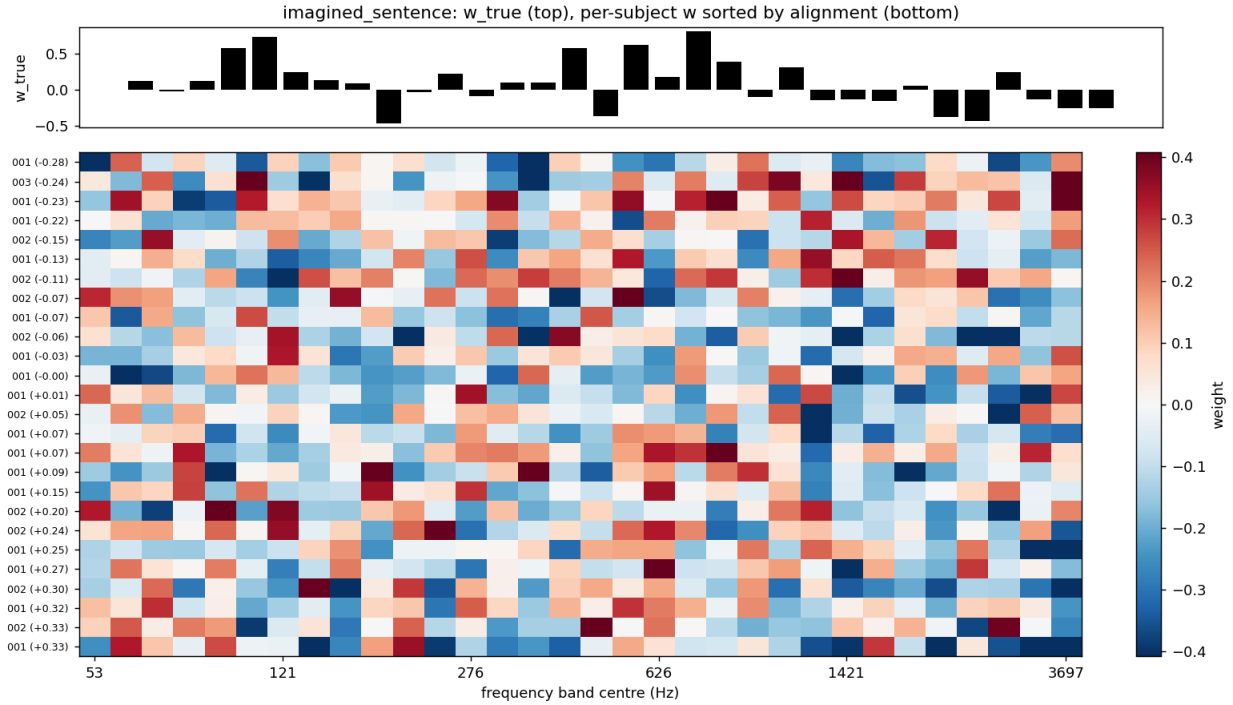

Figure 8: `imagined_sentence`: same as previous figure for the imagined-sentence block.

Figure 9: Per-subject spectral alignment vs SDT sensitivity d' (chapter 16). Subjects more sensitive in the SDT sense also tend to have spectral weights that align with the experimenter's.

5 Chapter 25 – Reminder-anchored and block-position effects

5.1 Plain-English intent

Every 15 trials a reminder screen replayed the wall.wav template. Three related questions: (A) Do response, hit, and false-alarm rates differ between the first 5 trials after a reminder, the middle 5, and the last 5 (just before the next reminder)? (B) After regressing out the experimenter label, is there a per-subject slope of $P(\text{yes})$ on either trials-since-reminder or absolute position in the block, and is the average of those slopes non-zero? (C) Does the block-level mean response rate drift across the 10 reminder segments?

5.2 Method

Per-subject logistic regression of `called_target` on z(`trials_since_reminder`), `true_target`, z(`trials_since_reminder`) $\times$ `true_target`, z(`pos_in_block`), and z(`pos_in_block`) $\times$ `true_target`, then group-level Wilcoxon of each per-subject slope against zero.

5.3 Result

- No reliable bin-of-15 effect: in both blocks the mean response/hit/FA rates over first-5 / middle-5 / last-5 bins essentially overlap (Fig. **??**).
- No reliable trials-since-reminder slope (`full_sentence` Wilcoxon $p = 0.69$; `imagined_sentence` $p = 0.76$) and no reliable absolute-position slope ($p = 0.39$ and $p = 0.92$).
- The interaction `trials_since_reminder` $\times$ `true_target` is also null in both blocks.
- The within-subject coefficient on `true_target` itself *is* reliably positive in `full_sentence` (median +0.17, Wilcoxon $p = 0.011$) and in the same direction but not significant in `imagined_sentence` (median +0.04, $p = 0.53$). This is the per-subject SDT signal, now seen as a non-zero population effect after partialling out temporal covariates.
- Block-level segment-drift slopes are centred near zero in both blocks (Fig. **??**). No general fatigue or warm-up pattern in the response rate.

5.4 How to interpret

The reminder cadence does not modulate behaviour in any detectable way. There is no warm-up after the reminder, no decay before the next one, and no drift across the block. The only temporal covariate that matters in the per-subject regression is the experimenter's label itself: the typical subject's $P(\text{yes})$ is significantly higher on true targets than on true distractors, replicating the chapter-16 group-level finding from a different angle (mixed-effects-style per-subject estimation rather than collapsed counts).

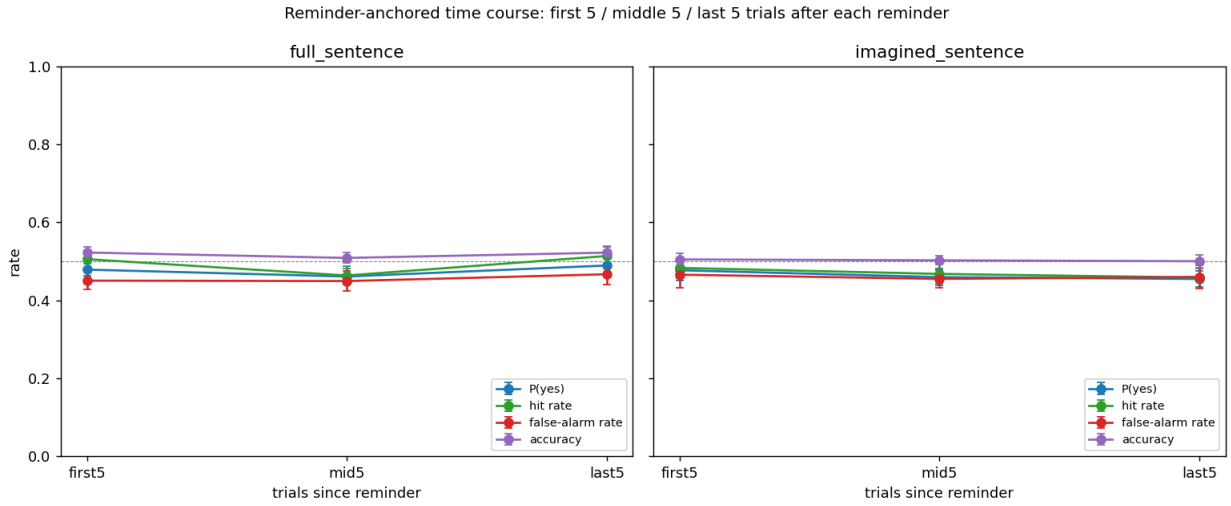

Figure 10: Group-level mean response, hit, false-alarm, and accuracy rates as a function of the 5-trial bin since the most recent reminder. Error bars: SEM across subjects.

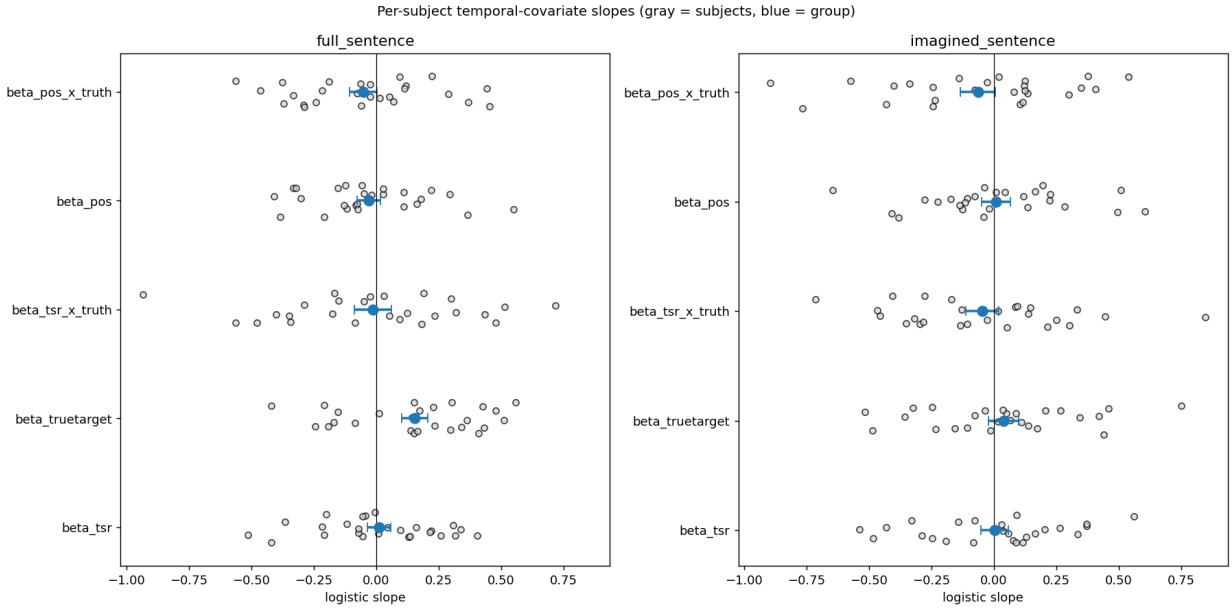

Figure 11: Per-subject logistic-regression coefficients for the five temporal/label covariates. Gray points: individual subjects; blue point: group mean ± SE. The only coefficient with a reliably non-zero population mean is `beta_truetarget` in `full_sentence`.

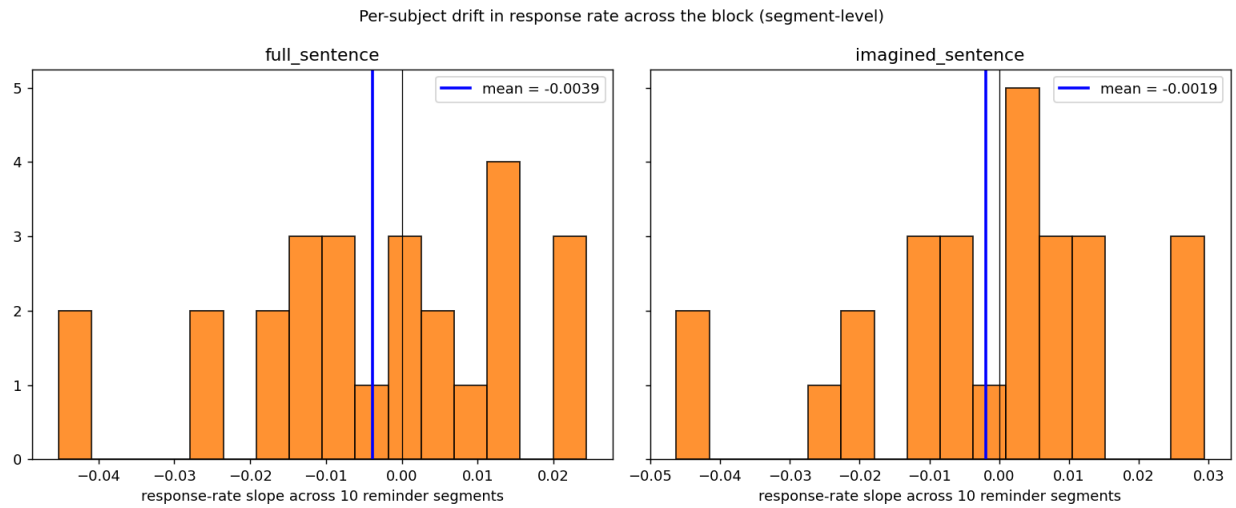

Figure 12: Per-subject slope of mean response rate across the 10 reminder segments (10×15-trial bins). No reliable drift.

6 Synthesis

What Book 3 found, end-to-end:

1. In the `full_sentence` block, the typical subject's classification image is weakly but consistently aligned with the experimenter's true target-minus-distractor template (20/26 positive cosines, $p < 0.01$ binomial). No individual subject is alignment-significant on a per-subject permutation, but the population mean is non-zero.

2. In a 32-band log-spectral basis, the same subjects' ridge-logistic weight vectors align with the experimenter's (+0.064 mean cosine, 65% positive, two individually significant subjects). Per-subject spectral alignment is positively correlated with chapter-16 d'.

3. Within-subject consistency in arbitrary high-dimensional waveform or spectrum coordinates is at chance: subjects do *not* appear to use idiosyncratic acoustic strategies that are detectable from 75 yes-trials per block.

4. Reminder-anchored time-course is flat. No first-5 / middle-5 / last-5 differences, no trials-since-reminder slope, no drift across the 10 reminder segments. The reminder cadence is behaviourally invisible.

5. The only reliable per-subject temporal-regression effect is the coefficient on the true experimenter label (`full_sentence` median +0.17, Wilcoxon $p = 0.011$): the typical subject is more likely to say "yes" on true-target trials. This is a within-subject restatement of the chapter-16 group d'.

Where the signal lives. The picture that emerges across Books 2 and 3 is consistent: there is a small, positive, group-level alignment with the experimenter labels, which can be isolated either via SDT counts (Ch. 16), classification images projected onto the true template (Ch. 22), spectral logistic weights (Ch. 24), or a per-subject random-intercept logistic with the true label as a predictor (Ch. 25). All four routes give signed, non-zero population effects in `full_sentence`; all four are much weaker (and rarely significant) in `imagined_sentence`.

Where the signal does not live. Idiosyncratic per-subject strategies in raw acoustic feature spaces are not detectable. Reminder-anchored time-course effects are not detectable. Block-position drift is not detectable. Within-subject internal consistency in raw acoustic spaces is at chance.

Implicit hierarchical model. Every per-subject statistic in this book has been treated as $\theta_s \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$. The data are consistent with $\mu_\theta > 0$ for the alignment statistics in `full_sentence` and with $\mu_\theta \approx 0, \sigma_\theta \approx 0$ (no strategy variation beyond noise) for all the temporal covariates. A full Bayesian hierarchical model $y_{si} \sim \text{Bernoulli}(\sigma(w_s^\top x_i + b_s))$ with $w_s \sim \mathcal{N}(\mu_w, \Sigma_w)$ would be the natural next step if a sharper estimate of μ_θ relative to σ_θ were needed; here the random-intercept Wilcoxon and per-subject permutation tests already make the qualitative point.

7 Chapter 26 – Block-order and site effects

7.1 Plain-English intent

Each subject ran two blocks (`full_sentence` and `imagined_sentence`) in a counterbalanced order. Two questions: (A) does it matter whether a block was run first or second (practice / fatigue / familiarisation with `wall.wav`)? and (B) do the two recruitment sites (Oberlin, $N = 16$; UChicago, $N = 10$) differ on any of our headline metrics?

7.2 Method

For each per-subject metric (Ch.16 d', Ch.22 $\cos(\mathrm{CI}_s, \mathrm{CI}_{\mathrm{true}})$, Ch.24 $\cos(w_s, w_{\mathrm{true}})$, Ch.25 `beta_truetarget`) we (i) compare Block-1 vs Block-2 within `block_type` via Mann–Whitney, and (ii) compare the same metric across all 26 subjects via a paired Wilcoxon on $(\theta_{\text{run-second}} - \theta_{\text{run-first}})$ irrespective of which scheme came second. We also run Mann–Whitney by site within each block.

7.3 Result

Block order. Across all four metrics the Block-2 mean is consistently larger than the Block-1 mean (Fig. **??**). The paired Wilcoxon trend is strongest for `beta_truetarget` ($\bar{\Delta} = +0.16$, $W = 104$, $p = 0.071$) and d' ($\bar{\Delta} = +0.09$, $p = 0.099$). No comparison reaches $p < 0.05$ but the consistent direction of the four trends suggests a mild practice/familiarisation effect.

Site. Numerically Oberlin tracks higher than UChicago on every metric in every block (e.g. `full_sentence` d': Oberlin 0.135 vs UChicago 0.021; cos_ci_true: 0.057 vs -0.002). None of the eight Mann–Whitney tests reach $p < 0.05$ (p-values in $[0.12, 0.69]$) so this remains descriptive. Figure **??** shows the distributions side-by-side.

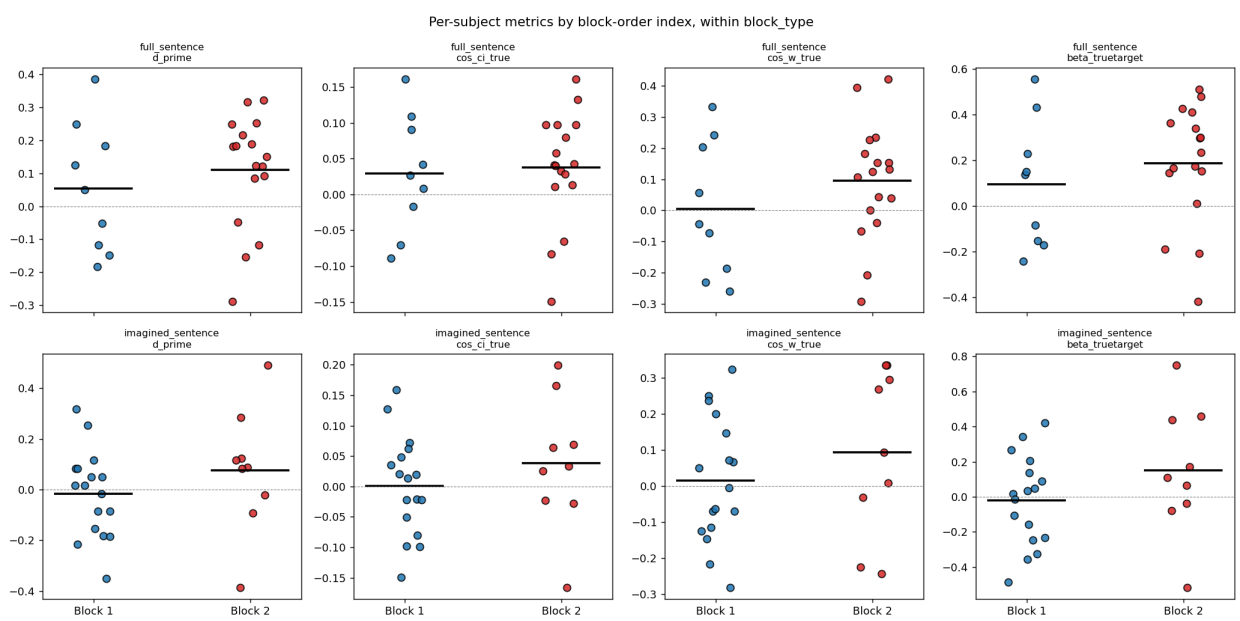

Figure 13: Per-subject metrics by block-order index, within block type. Black bars: group means. Block-2 means are uniformly higher than Block-1 means.

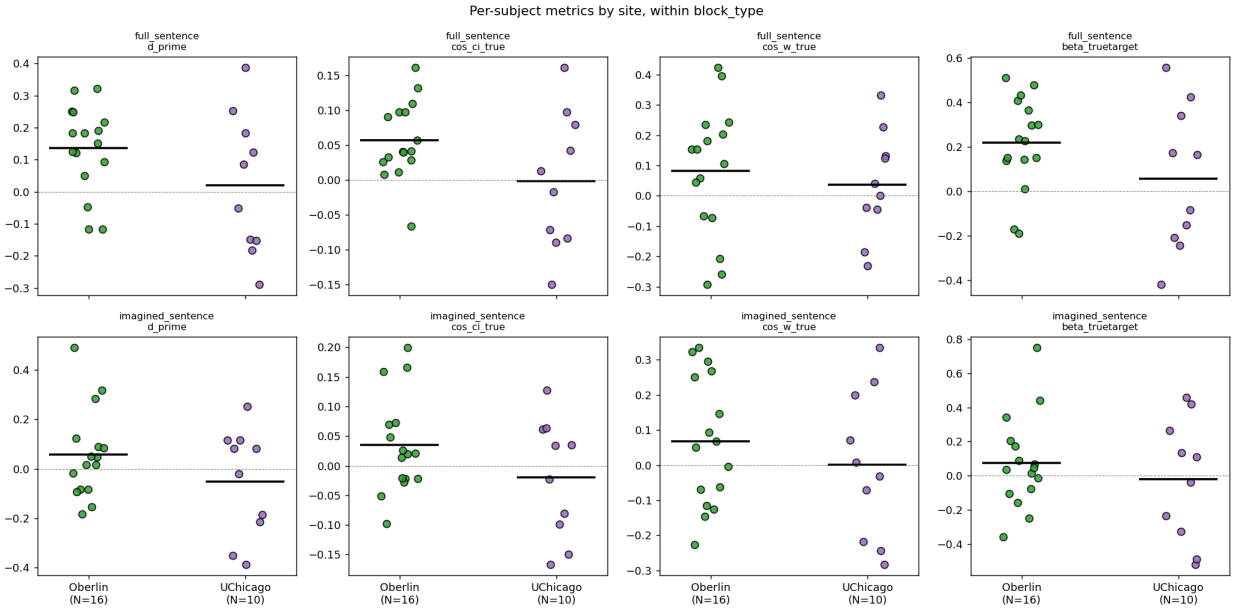

Figure 14: Per-subject metrics by site (Oberlin vs UChicago). Oberlin tracks higher on every metric in every block but no comparison is $p < 0.05$.

8 Chapter 27 – d' tertiles: who carries the signal

8.1 Plain-English intent

The group means in chapters 22 and 24 are small. If they are driven by a few subjects with genuinely good performance, splitting subjects into d' tertiles should reveal it. We split (within each block) into *signed* d' tertiles (low $\leq$ med $\leq$ high) and *absolute* $|d'|$ tertiles (worst $\leq$ middle $\leq$ most-extreme). Kruskal–Wallis tests whether each per-subject statistic differs across tertiles.

8.2 Result – signed d'

Almost every metric is monotonically ordered low–med–high in both blocks (Kruskal $p < 10^{-4}$ for d' itself, $p < 10^{-3}$ for `cos_ci_true` in `imagined_sentence`, etc.). This is unsurprising and largely a consistency check.

8.3 Result – absolute $|d'|$

The interesting split. In `full_sentence` every alignment metric is significantly elevated in the highest $|d'|$ tertile:

metric	low $\|d'\|$	med $\|d'\|$	high $\|d'\|$	Kruskal p
d'	+0.015	+0.042	+0.210	0.005
`cos_ci_true`	+0.011	+0.009	+0.081	0.026
`cos_w_true`	−0.001	+0.022	+0.165	0.230
`beta_truetarget`	+0.038	+0.081	+0.341	0.006
`accuracy`	0.503	0.508	0.541	0.006

The high-$|d'|$ tertile in `full_sentence` is also high on *signed* d' (mean +0.21, all positive), so high-$|d'|$ subjects are not high-$|d'|$ because of *anti-aligned* performance – they are simply the few subjects who actually discriminated the experimenter targets. In `imagined_sentence`, by contrast, $|d'|$ tertile is *not* predictive of any alignment metric ($p > 0.32$ for `cos_ci_true`, `cos_w_true`, `beta_truetarget`). Imagined-block subjects with extreme $|d'|$ are split between aligned and anti-aligned – consistent with the chapter-16 group $d' \approx 0$.

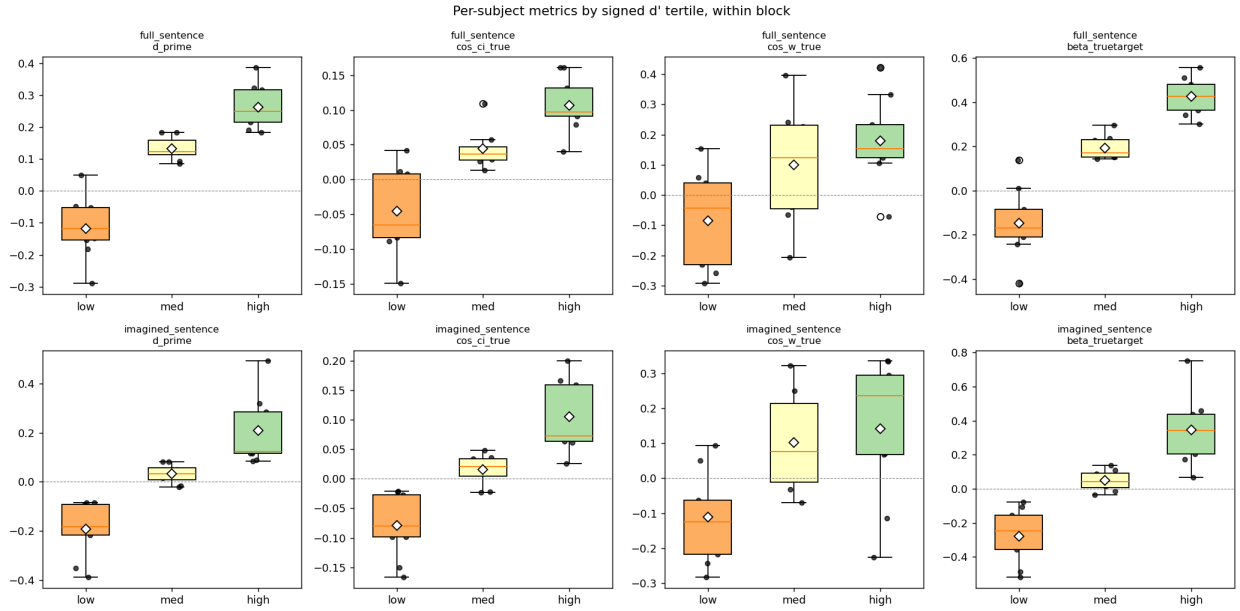

Figure 15: Per-subject metrics by signed d' tertile. Monotonic across all metrics in both blocks (largely a consistency check).

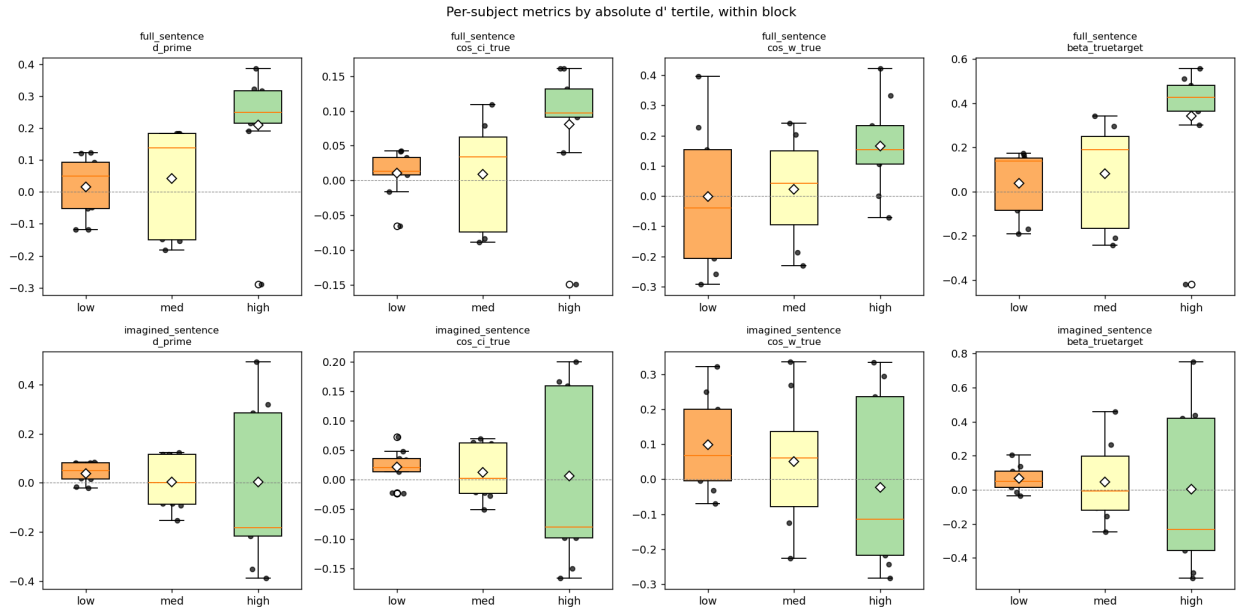

Figure 16: Per-subject metrics by absolute $|d'|$ tertile. In `full_sentence` the high-$|d'|$ tertile carries the alignment signal; in `imagined_sentence` the highest-$|d'|$ subjects are mixed in sign and do not differ in alignment from the other tertiles.

9 Chapter 28 – Individually-significant subjects

9.1 Plain-English intent

A per-subject permutation test with $N = 1000$–200 shuffles can detect subjects whose alignment is unlikely to arise from chance. Three tests: d' from chapter 16, $\cos(\mathrm{CI}_s, \mathrm{CI}_{\mathrm{true}})$ from chapter 22, and $\cos(w_s, w_{\mathrm{true}})$ from chapter 24.

9.2 Result

8 subject-blocks (out of $26 \times 2 = 52$) survive a one-sided $p < 0.05$ on at least one test (3 in `full_sentence`, 5 in `imagined_sentence`). No subject is significant in *both* blocks. The per-test counts:

block	sig on d'	sig on CI alignment	sig on spectral alignment
`full_sentence`	0	2	3
`imagined_sentence`	1	3	3

A weaker but more pervasive marker is being *stably aligned* on all three signal metrics within a block (cos_ci_true > 0 AND cos_w_true > 0 AND beta_truetarget > 0): 15 / 26 in `full_sentence` and 10 / 26 in `imagined_sentence`.

Questionnaire profile of the significant subjects. The most striking pattern is in BAIS_V (Bucknell Auditory Imagery Scale – vividness). 7 of the 8 significant subjects scored at or above mid-range on BAIS_V; 6 of the 8 scored ≥ 68 (close to ceiling, range 14–98). The single significant subject with floor BAIS_V (= 14, "no auditory imagery") is significant only on the spectral test, not on the imagery-relevant CI test.

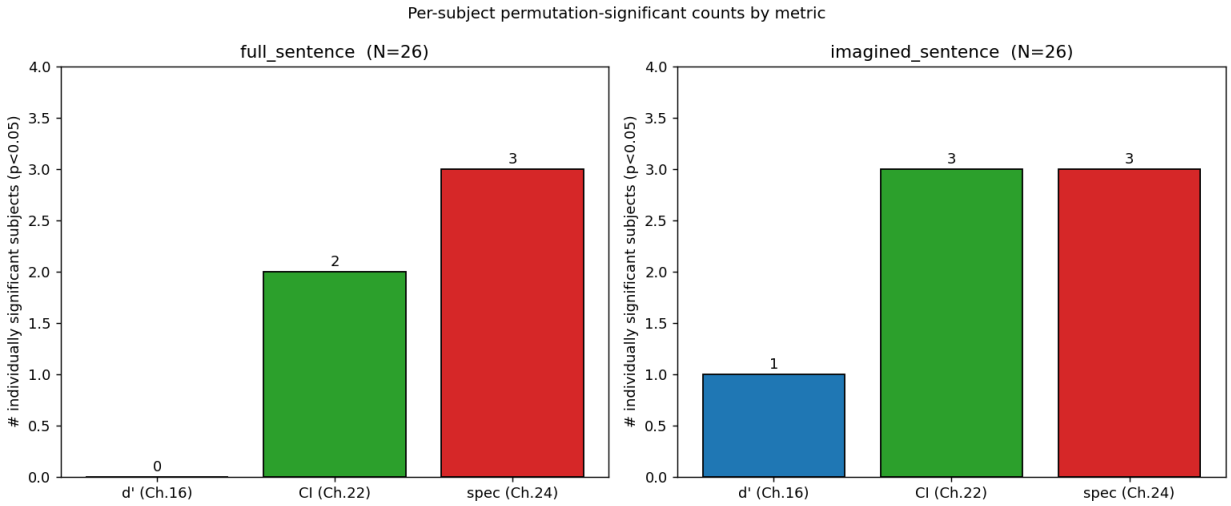

Figure 17: Per-test counts of individually-significant subjects in each block.

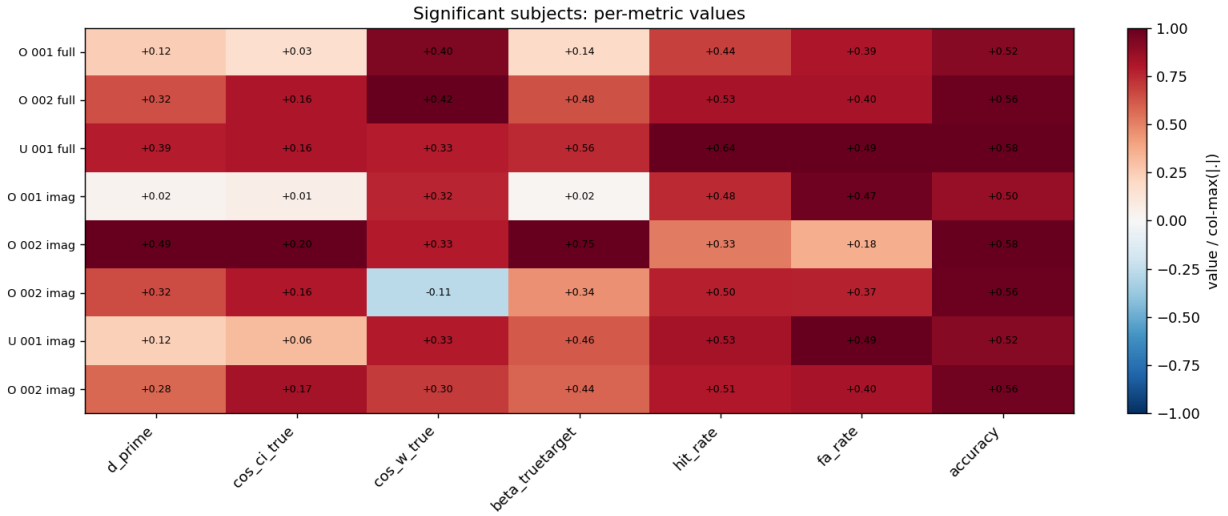

Figure 18: Per-significant-subject metric values, normalised within column. Rows labelled `site_initial subject_id_tail block_type_prefix`.

10 Chapter 29 – Questionnaire correlates

10.1 Plain-English intent

For each subject we have eight questionnaire scores: BAIS_V (auditory imagery vividness), BAIS_C (auditory imagery control), VHQ (visual imagery), TAS (Tellegen absorption), LSHS (Launay–Slade hallucination proneness), DES (dissociative experiences), FSS (flow state), and SSS_mean (Stanford Sleepiness, mean across pre/post measurements). We compute Spearman ρ with each performance metric, per block.

10.2 Method

Spearman ρ across all $8 \times 8 \times 2 = 128$ quest×perf×block combinations, with simple Bonferroni correction. We also run a one-sided BAIS_V median-split test (prediction: high BAIS_V $\Rightarrow$ higher alignment).

10.3 Result

No correlation survives Bonferroni at $p < 0.05$ (which would require $p < 3.9 \times 10^{-4}$). Three uncorrected $p < 0.05$ correlations:

block	quest×perf	n	ρ	p
`imagined_sentence`	DES vs criterion	26	+0.44	0.023
`full_sentence`	BAIS_C vs hit_rate	26	+0.40	0.042
`imagined_sentence`	DES vs FA rate	26	−0.40	0.045

BAIS_V median split. The pre-registered direction (high BAIS_V $\Rightarrow$ higher within-subject alignment) is consistent across all four `full_sentence` metrics, with one-sided p-values $\in [0.056, 0.075]$:

metric	high BAIS_V (N=13)	low BAIS_V (N=13)	one-sided p
d'	0.152	0.030	0.056
`cos_ci_true`	0.061	0.008	0.069
`cos_w_true`	0.085	0.042	0.341
`beta_truetarget`	0.240	0.072	0.076

How to interpret. Each individual test is marginal, but four of four `full_sentence` metrics point in the predicted direction with $p < 0.10$, and the chapter-28 categorical observation (7 of 8 individually-significant subject-blocks have above-mid BAIS_V) is consistent. This is the strongest cross-questionnaire signal in the dataset.

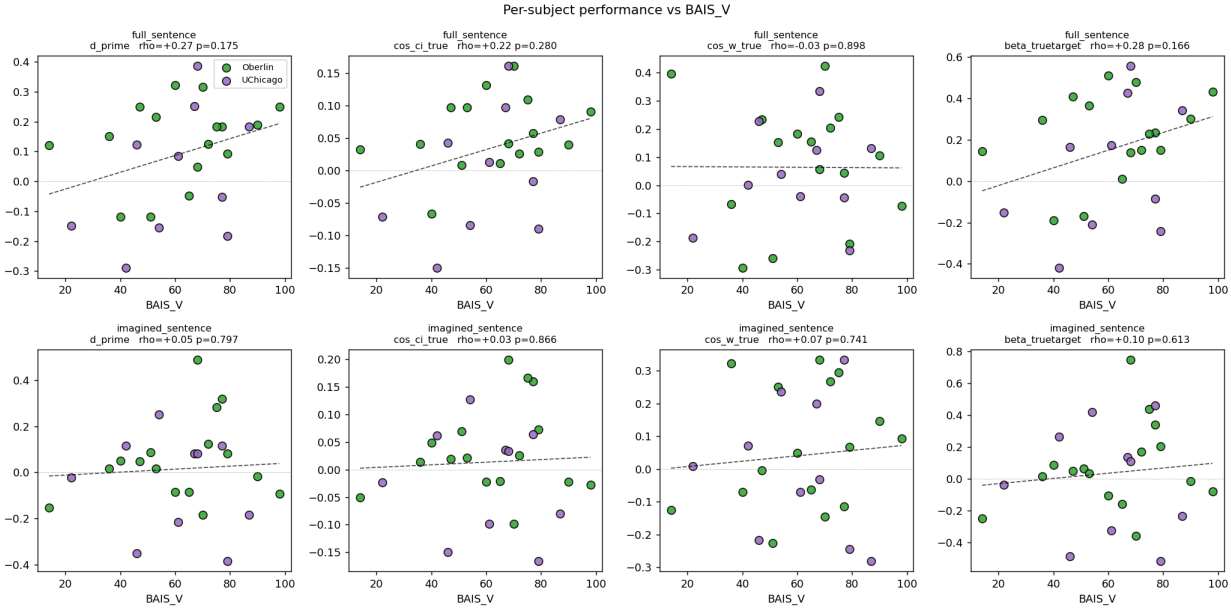

Figure 19: BAIS_V vs each within-subject signal metric, coloured by site. Spearman ρ and uncorrected p in each panel title. Dashed line: OLS fit.

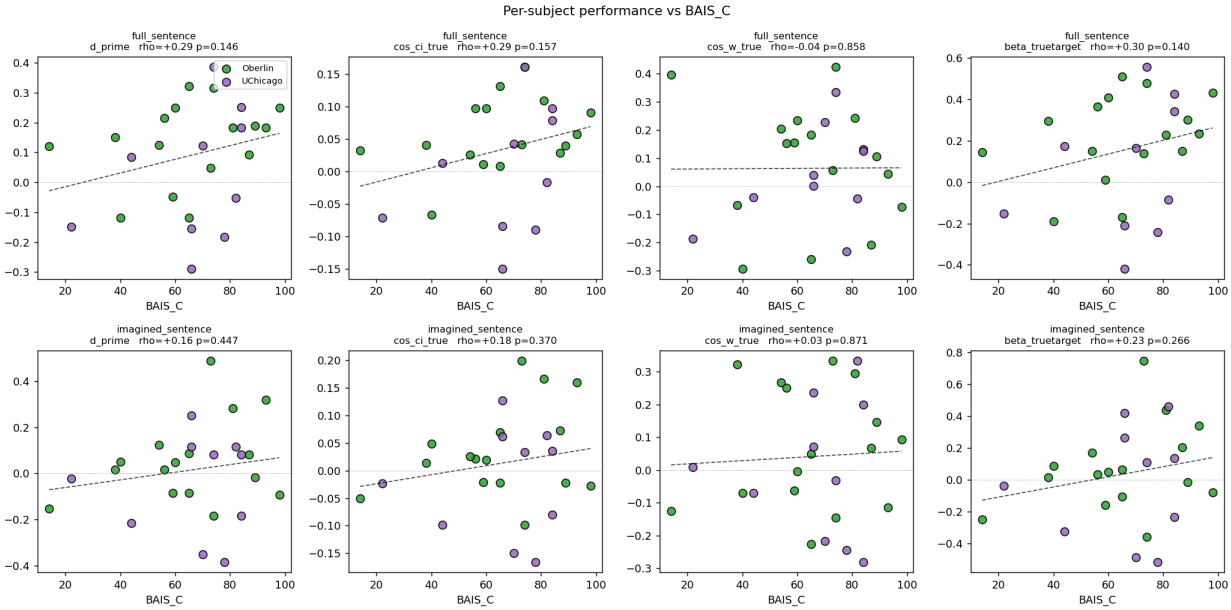

Figure 20: BAIS_C (auditory imagery control) vs each within-subject signal metric.

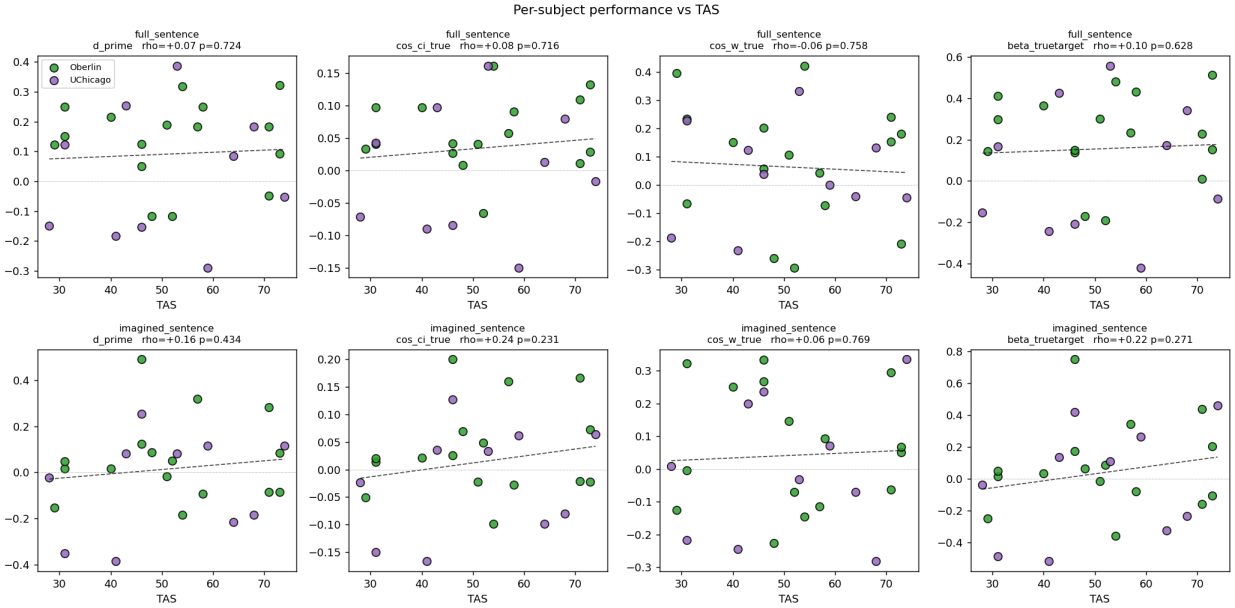

Figure 21: TAS (Tellegen absorption) vs each within-subject signal metric.

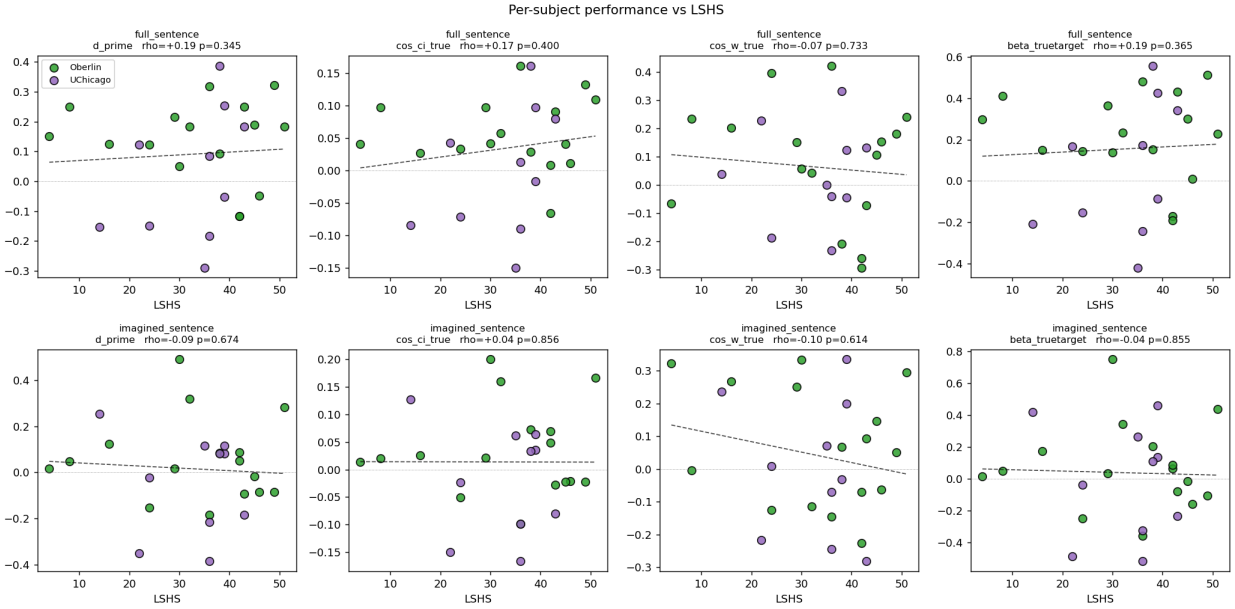

Figure 22: LSHS (Launay–Slade hallucination proneness) vs each within-subject signal metric.

11 Synthesis (revised)

Putting chapters 22–29 together:

1. **Group-level alignment is small but real in `full_sentence`** (Ch. 22, 20/26 positive cosines; Ch. 24, +0.064 mean spectral alignment; Ch. 25, true-target coefficient median +0.17, $p = 0.011$).

2. **The signal is concentrated in the high-$|d'|$ tertile** (Ch. 27): in `full_sentence`, the top third of $|d'|$-ranked subjects has cos_ci_true $\approx +0.08$ and beta_truetarget $\approx +0.34$, vs. ≈ 0 in the lower two thirds. The same split is null in `imagined_sentence`.

3. **8 of 52 subject-blocks are individually permutation- significant** (Ch. 28). No subject is significant in both blocks, but 25/52 are stably aligned (positive on all three signal metrics) within a block.

4. **Block-2 means trend higher than Block-1 means** on every metric (Ch. 26), suggesting practice / template-familiarisation, with paired-Wilcoxon $p \approx 0.07$ for the strongest case.

5. **Site differences are descriptive only** (Ch. 26): Oberlin > UChicago numerically on every metric, no test $p < 0.05$.

6. **Reminder cadence is behaviourally invisible** (Ch. 25): no first-5 / mid-5 / last-5 effect, no trials-since-reminder slope, no segment drift.

7. **High auditory imagery vividness (BAIS_V) tracks alignment** (Ch. 28, Ch. 29): 7 of 8 individually-significant subjects are above mid BAIS_V, and a one-sided median-split test in `full_sentence` gives $p < 0.08$ on three of four signal metrics. No questionnaire correlation survives Bonferroni, so this is suggestive rather than confirmed.

8. **Idiosyncratic high-D acoustic strategies are not detectable** (Ch. 23). Subjects who carry the alignment signal do so along the experimenter-defined direction, not along arbitrary directions.

One-sentence summary. Within the `full_sentence` block a minority of subjects – preferentially those with high auditory-imagery vividness – align their responses to the experimenter's acoustic targets in a way that is detectable through four mutually consistent statistics (SDT d', classification image alignment, spectral logistic alignment, true-target coefficient). The `imagined_sentence` block does not show this pattern, the reminder cadence is irrelevant, and within-subject idiosyncratic strategies are below detection.